SPECIFICATION TITLE	SPEC NUMBER	REVISION	EFFECTIVE DATE
CHARPY IMPACT TESTING	**CMS-004**	**C**	**08/15/2019**

Scope	To establish the Cactus Wellhead requirements for charpy impact testing of forgings, castings, wrought products such as bars and mill shapes and weld qualifications. Suppliers shall use this specification in conjunction with a specific Cactus Wellhead Material Specification (CMS) or supporting Engineering Product Specification Report (EPS).
Reference Standards	• API Specification 6A, Specification of Wellhead and Tree Equipment • ASTM A370, Standard Test Methods and Definitions for Mechanical Testing of Steel Products
Test Specimens	Impact test specimens shall be removed from a Qualification Test Coupon (QTC) as described in one of the following CMS's, as applicable. o Cactus Material Specification CMS-001, Material Requirements for Ferrous Castings; or o Cactus Material Specification CMS-002, Material Requirements for Forgings, Bars and Mill Shapes Impact tests of a QTC shall be used to qualify a heat and the bodies, bonnets, stems, and end and outlet connectors produced from that heat and shall conform to the requirements of Table 1.
Impact Strength Requirements	• Impact values for standard and non-standard materials shall conform to the requirements of Table 1 or Table 2. • If sub-size specimens are used, the Charpy V-notch impact requirements shall be equal to that of the 10 mm x 10 mm specimens multiplied by the adjustment factor listed in Table 2.
Test Methods	• Perform impact tests in accordance with the procedures specified in ASTM A370 using the Charpy V-notch technique as described in Cactus Material Specification CMS-001 or CMS-002. • To quality a material for a temperature range, perform the impact tests at or below the lowest temperature of the classification range, in order to qualify a material for a temperature rating. • Test a minimum of three impact specimens to qualify a heat of material. • Impact properties as determined from these tests shall satisfy Table 1 or Table 2. • In no case shall an individual impact value fall below two-thirds of that required as a minimum average.

	• No more than one specimen shall fall below the required minimum average. • If a test fails, then a retest of three additional specimens (typically removed from the same location within the same QTC with no additional heat treatment) may be made, each of which shall exhibit an impact value equal to or exceeding the required minimum average value.
Specimen Orientation	• The values listed in Table 1 and Table 2 are the minimum acceptable values for wrought products, castings and weld qualifications tested in the transverse direction. • Wrought products may be tested in the longitudinal direction instead of the transverse direction and shall meet the requirements of Table 1 and Table 2. • Castings have no directionality; the values of the transverse direction of Table 1 and Table 2 apply.
Applicable Cactus Wellhead Components and Parts Requiring Charpy Impacts	Charpy Impact testing is applicable to the following Cactus Wellhead PSL 1 and PSL 2 components and parts starting at Temperature Class P and PSL 3 starting at Temperature Class U/V. CMS-004 will appear in the EPS bill of material(s) or item description when charpy impact testing is required. • Valve bodies, wellhead bodies, crosses, tees; • Bonnets and stems; • End and outlet connectors such as: flanges, cross-over connectors, TA caps, tree caps; • Mandrel Type Hangers; • Bull-plugs, back-pressure valve bodies and valve-removal plug bodies (no PSL); • Actuator DAH model series: bonnet adapter, coupler, piston, rod indicator, cylinder, retainer dog, end cap (no PSL); • Nonintegral Metal Seals such as MCS secondary seals (no PSL); and • Studs, bolts and screws for Temperature Class K/L and those larger than 2.50” thread size regardless of temperature class.

Table 1– Charpy V-notch Impact Requirements – 10 mm x 10 mm

Temperature ◦F (◦C)		Minimum Average Impact Value Transverse Direction Ft-Lb (J)		
Classification	**Test**	**PSL 1**	**PSL 2**	**PSL 3**
K	-75 (-60)	15 (20)		15 (20)
L	-50 (-46)	15 (20)		15 (20)
N	-50 (-46)	15 (20)		15 (20)
P	-20 (-29)	**15 (20)**		15 (20)
S	0 (-18)	---		15 (20)
T	0 (-18)	---		15 (20)
U	0 (-18)	---		15 (20)
V	0 (-18)	---		15 (20)
Temperature ◦F (◦C)		**Minimum Average Impact Value Longitudinal Direction Ft-Lb (J)**		
Classification	**Test**	**PSL 1**	**PSL 2**	**PSL 3**
K	-75 (-60)	20 (27)		20 (27)
L	-50 (-46)	20 (27)		20 (27)
N	-50 (-46)	20 (27)		20 (27)
P	-20 (-29)	20 (27)		20 (27)
S	0 (-18)	---		20 (27)
T	0 (-18)	---		20 (27)
U	0 (-18)	---		20 (27)
V	0 (-18)	---		20 (27)

Table 2 – Adjustment Factors for Sub-Size Impact Specimens (PSL 1 to PSL 3)

Specimen Dimensions	Adjustment Factor	Minimum Average Impact Value Wrought Materials	
		Transverse Direction Wrought and Castings Ft-Lb (J)	Longitudinal Direction Wrought Ft-Lb (J)
		PSL 1 to PSL 3	PSL 1 to PSL 3
10 mm x 10 mm (full size)	1 (none)	15 (20)	20 (27)
10 mm x 7.5 mm	0.833	13 (17)	17 (23)
10 mm x 6.7 mm	0.780	12 (16)	16 (21)
10 mm x 5.0 mm	0.667	10 (13)	13 (18)
10 mm x 3.3 mm	0.440	7 (9)	9 (12)
10 mm x 2.5 mm	0.333	5 (7)	7 (9)

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Date Prepared	*July 10, 2019*	***Date Approved:***	*July 10, 2019*

Revision History Table		
C	07/10/2019	Updated to address compliance with API 6A, 21st edition, Specification for Wellhead and Tree Equipment, effective November 1, 2019. Updated references, test specimens, added applicable Cactus Wellhead product. Added charpy impact requirements for Temp P, PSL 1. Added 10mm X 10mm longitudinal charpy impact values.
B	1/12/2012	Editorial correction on test temperature
A	9/1/2011	Initial release